

LEVEL 4 – HC VERMA / TOUGH MAINS JEE QUESTIONS

1. Basics of Electrochemistry

1. A solution contains 0.01 M KCl. The measured conductivity is $1.41 \times 10^{-3} \text{ S}\cdot\text{cm}^{-1}$. Calculate molar conductivity and degree of dissociation if $\lambda_m^0 = 149 \text{ S}\cdot\text{cm}^2\cdot\text{mol}^{-1}$.
 2. The conductivity of 0.01 M acetic acid is $0.0004 \text{ S}\cdot\text{cm}^{-1}$. $\lambda_m^0 = 390 \text{ S}\cdot\text{cm}^2\cdot\text{mol}^{-1}$. Find degree of dissociation and concentration of ions.
 3. Molar conductivity of a 0.1 M strong electrolyte decreases with concentration. Explain mathematically using Kohlrausch's law.
 4. Two solutions of NaCl (0.001 M and 0.01 M) are mixed. Calculate final conductivity, assuming additivity.
 5. Ionic mobility of $\text{Na}^+ = 5 \times 10^{-8} \text{ m}^2\cdot\text{V}^{-1}\cdot\text{s}^{-1}$, $\text{Cl}^- = 7.9 \times 10^{-8} \text{ m}^2\cdot\text{V}^{-1}\cdot\text{s}^{-1}$. Calculate limiting molar conductivity.
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2. Electrolytic and Galvanic Cells

6. A Zn-Cu cell has $[\text{Zn}^{2+}] = 0.01 \text{ M}$ and $[\text{Cu}^{2+}] = 0.1 \text{ M}$. Calculate EMF at 298 K.
 7. Electrolysis of 1 M CuSO_4 solution using 2 A current for 30 minutes. Calculate mass of Cu deposited.
 8. 96500 C deposit 107.9 g of Ag. How much mass is deposited by 1 F?
 9. Daniell cell: $\text{Zn}^{2+} = 0.1 \text{ M}$, $\text{Cu}^{2+} = 0.01 \text{ M}$. Which electrode potential changes faster during discharge?
 10. An electrolytic cell is used to deposit 10 g of Al. Current = 5 A. Calculate time required (Al = 27, $n = 3$).
 11. Electrolysis of molten NaCl produces 11.6 g of Na. Calculate total charge passed.
 12. A cell: $\text{Fe}/\text{Fe}^{2+} \parallel \text{Cu}^{2+}/\text{Cu}$. $[\text{Fe}^{2+}] = 0.01 \text{ M}$, $[\text{Cu}^{2+}] = 0.1 \text{ M}$. Calculate EMF using Nernst equation.
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3. Electrode Potential & Nernst Applications

13. Standard potentials: $\text{Zn}^{2+}/\text{Zn} = -0.76 \text{ V}$, $\text{Cu}^{2+}/\text{Cu} = +0.34 \text{ V}$. Calculate E_{cell} at $[\text{Zn}^{2+}] = 0.01 \text{ M}$, $[\text{Cu}^{2+}] = 0.001 \text{ M}$.
14. $2\text{Ag}^+ + \text{Cu} \rightarrow 2\text{Ag} + \text{Cu}^{2+}$. Find EMF when $[\text{Ag}^+] = 0.01 \text{ M}$, $[\text{Cu}^{2+}] = 0.1 \text{ M}$ at 298 K.
15. Electrode potential of $\text{Zn}^{2+}/\text{Zn} = -0.76 \text{ V}$. At what $[\text{Zn}^{2+}]$ does potential become -0.66 V ?
16. If cell EMF = 0, calculate equilibrium constant K for reaction $\text{Fe}^{2+} + \text{Cu} \rightarrow \text{Fe} + \text{Cu}^{2+}$.
17. Predict direction of reaction: $\text{Fe} + \text{Cu}^{2+} \rightarrow \text{Fe}^{2+} + \text{Cu}$.
18. Standard EMF of cell = 1.10 V. At non-standard concentrations, EMF = 1.05 V. Calculate reaction quotient Q.
19. Nernst equation: Explain effect of decreasing reactant concentration on EMF.
20. Cell reaction: $\text{Zn} + 2\text{Ag}^+ \rightarrow \text{Zn}^{2+} + 2\text{Ag}$. Determine EMF if $[\text{Ag}^+] = 0.01 \text{ M}$, $[\text{Zn}^{2+}] = 0.1 \text{ M}$.
21. For Cu^{2+}/Cu electrode, $[\text{Cu}^{2+}] = 10^{-4} \text{ M}$. Find potential at 298 K.

4. Electrochemical Series & Corrosion

22. Rank metals: Mg, Al, Zn, Fe, Cu by corrosion tendency.
 23. Sacrificial anode: Steel protected using Zn. Explain which is cathode and which corrodes.
 24. Predict if Fe will displace Cu^{2+} from CuSO_4 solution.
 25. Rusting of iron accelerated by:
 - a) Acidic environment
 - b) Presence of NaCl
 - c) Oxygen availability
 - d) All
 26. Calculate EMF of cell $\text{Fe}/\text{Fe}^{2+} \parallel \text{Cu}^{2+}/\text{Cu}$ to predict corrosion.
 27. Electrochemical series predicts: Which metal is the strongest oxidizing agent?
 28. Cathodic protection applied to underground pipes: Why zinc corrodes first?
 29. Mg placed in ZnSO_4 solution. Reaction occurs? Predict products.
 30. Explain why aluminum forms protective oxide layer but iron rusts.
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5. Faraday's Laws & Electrolysis Calculations

31. How many grams of Cu are deposited by 193000 C of electricity?
 32. Electrolysis of water produces 11.2 L H_2 . Calculate charge required.
 33. 2 F of electricity deposit 107.9 g Ag. Equivalent weight?
 34. Current = 2 A, time = 1 hour. Mass of Cu deposited?
 35. Electroplating: 5 g Cu deposited. Current = 2 A. Calculate deposition time.
 36. Electrolysis of molten NaCl produces 11.6 g Na. Find total charge.
 37. How many moles of electrons required to deposit 1 mole of Al^{3+} ?
 38. Calculate Faraday constant from given data: 1 F deposits 107.9 g Ag.
 39. Electrorefining of Cu: 100 g impure Cu anode. Mass of pure Cu deposited?
 40. Mass deposited $\propto$: Current, time, equivalent weight. Show calculation numerically.
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6. Batteries & Fuel Cells

41. Li-ion battery: Charge = 2 Ah, Voltage = 3.7 V. Calculate energy stored.
42. Lead-acid battery: 12 V, 10 A, 2 hours. Energy delivered?
43. Ni-Cd battery: 1.2 V, capacity 2 Ah. Energy?
44. Fuel cell: 1 mole H_2 reacts with O_2 . Calculate electrical energy produced ($E = 1.23$ V, $F = 96500$ C).
45. 100 H_2 - O_2 fuel cells in series, each 1.23 V. Total voltage?
46. Battery reactions: Balance electrons in Ni-Cd battery discharge.
47. Why Li-ion preferred over Ni-Cd for mobile devices?
48. Calculate total charge in 5 Ah Li-ion battery.
49. Electric vehicle uses 40 kWh. How many Li-ion cells needed (each 3.7 V, 2 Ah)?

50. H_2 fuel cell: Determine work done per mole of H_2 consumed.

7. Conductometric, Potentiometric & Voltammetric Applications

- 51. Conductometric titration: weak acid + strong base. Explain conductivity minimum at equivalence point.
 - 52. Potentiometric titration: EMF vs volume graph, identify endpoint.
 - 53. Voltammetry: Determine concentration of 10^{-6} M Pb^{2+} using current-peak method.
 - 54. Limiting molar conductivity calculation for strong electrolyte using Kohlrausch's law.
 - 55. Conductometric titration: Explain effect of dilution on titration curve.
 - 56. Potentiometric titration: Determine unknown concentration from EMF measurements.
 - 57. Voltammetry: Explain effect of scan rate on peak current.
 - 58. Conductivity unit conversion: $\text{S}\cdot\text{cm}^{-1} \rightarrow \text{S}\cdot\text{m}^{-1}$.
 - 59. Determine degree of dissociation from molar conductivity data.
 - 60. Explain why conductometric titration useful for weak acid-strong base titrations.
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8. Applications of Electrochemistry

- 61. Electroplating 10 g of Ag using 2 A current. Calculate time required.
- 62. Electrorefining: 100 g impure Cu anode. Calculate pure Cu deposited using 1 F.
- 63. Chlor-alkali process: Calculate moles of Cl_2 produced by 2 F of electricity.
- 64. Electrolysis of Al_2O_3 : 1 mole Al requires how many C of charge?
- 65. Fuel cells: 1 mole H_2 produces 2 F of electrons. Calculate energy produced.
- 66. Electroplating thickness: 5 g Cu on 100 cm^2 plate. Calculate thickness (density Cu = 8.96 g/cm^3).
- 67. Industrial electrochemistry: Explain why NaOH is produced in brine electrolysis.
- 68. Voltammetry: Determine peak current proportionality with concentration.
- 69. Potentiometric redox titration: Find unknown Fe^{2+} concentration using EMF readings.
- 70. Electrochemical methods in environmental analysis: Detect heavy metals using voltammetry.